Journal of Environmental Management 401 (2026) 128949

Contents lists available at ScienceDirect

Journal of Environmental Management

journal homepage: www.elsevier.com/locate/jenvman

Review

AI for smart wastewater treatment plants: A review of physics-informed water quality modeling, optimization, and advanced control

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ARTICLE INFO

Keywords:
Prediction
Optimization
Control
Physics-informed
Closed loop framework

ABSTRACT

Wastewater treatment plants (WWTPs) are among the largest energy consumers in cities and are also an important source of global greenhouse gas emissions. The use of artificial intelligence (AI) offers a promising route to make WWTPs low-carbon and smart. This review summarizes research progress over the past decade on AI for effluent quality prediction, process optimization, and advanced control in WWTPs, with an emphasis on how these methods support the development of smart WWTPs. The literature shows that machine learning and deep learning are widely used to predict key effluent indicators. When combined with multi-objective optimization, AI can balance effluent quality, energy use, and carbon emissions, and model-based and reinforcement learning control can further unlock energy saving and emission reduction potential. However, current applications still face limited interpretability, weak transferability across plants, and poor robustness to changing operating conditions. Therefore, this review focuses on physics-informed and other hybrid modeling approaches that embed activated sludge models and process constraints into neural networks, and discusses pathways to couple these models with real-time optimization and control frameworks. Finally, future research needs are identified in data infrastructure, interpretable decision support tools, and full-scale validation studies to enable trustworthy AI-enabled WWTPs and accelerate sustainable and intelligent transformation of the sector.

1. Introduction

With ongoing global industrialization and urbanization, water scarcity and water pollution are becoming more severe. According to the World Resources Institute, global greenhouse gas emissions reached 59 Gt CO_2-eq in 2024, of which wastewater treatment accounted for 1.3% (Li et al., 2025b). In 2024, 44% of wastewater was not treated or reused (Fig. 1). Meanwhile, taking the data of WWTPs in China as an example, Fig. 1 shows that emissions have remained high, while electricity consumption has increased year by year. This indicates that conventional mechanistic models, such as the activated sludge model, have clear limits when dealing with the nonlinear dynamics, long time delays, strong coupling, and disturbances in wastewater treatment processes, leading to high emissions and inefficient energy use. These issues reduce overall treatment efficiency (Chen et al., 2021) and hinder the transition of WWTPs toward intelligent operation.

With the rise of Industry 4.0 and smart water management, the wastewater treatment sector is moving toward intelligent and sustainable operation. The integration of the Internet of Things, AI, big data analytics, and digital twins provides key support for building a new generation of smart water treatment systems (Kurniawan et al., 2024), as shown in Fig. 2. Among these technologies, AI is widely seen as an effective tool to address the limitations of conventional methods. Researchers have developed AI applications for wastewater treatment, including water quality prediction, optimization of operating strategies, and adaptive control, to reduce energy consumption and carbon emissions (Li et al., 2021b).

In the intelligent transformation of WWTPs, water quality prediction and process control optimization are two key technical pillars. AI-based water quality prediction models can capture the nonlinear dynamics of treatment processes and, together with real-time monitoring data, forecast water quality changes. This enables early detection of abnormal

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https://doi.org/10.1016/j.jenvman.2026.128949
Received 7 January 2026; Received in revised form 31 January 2026; Accepted 6 February 2026
Available online 10 February 2026

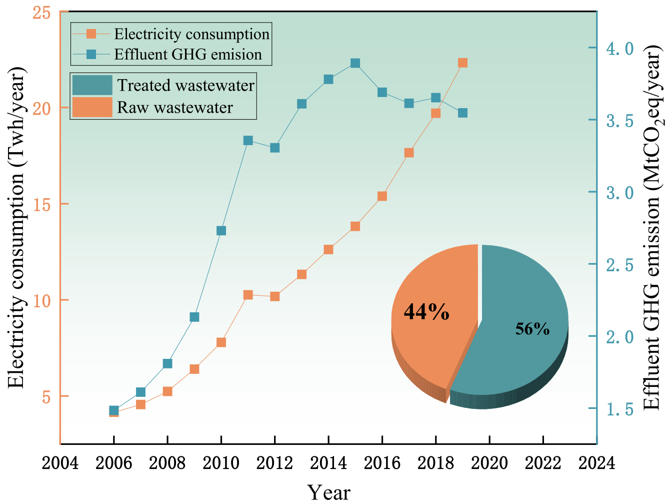

Fig. 1. Interannual statistics of carbon emissions and energy consumption in WWTPs in China.

operation and supports timely adjustments of aeration rate, external carbon dosing, and return sludge ratio (Wang et al., 2025a). Such data-driven models learn from large historical datasets to support parameter tuning and water quality prediction, avoiding the need to calibrate many hard-to-measure parameters required by conventional activated sludge models (ASMs), and thereby improving timeliness and prediction accuracy (Dairi et al., 2019). AI-based process control optimization can use intelligent optimization algorithms to automatically identify optimal setpoints, and then apply advanced control to adjust dissolved oxygen setpoints, carbon dosing, and related actions in advance, ensuring stable and efficient plant operation. Advanced control and multi-objective optimization allow setpoints and dosing strategies to be updated proactively based on predictions, while accounting for multiple objectives in WWTP operation, including effluent quality, energy use, chemical consumption, carbon emissions, and operating cost. Multi-objective optimal control aims to achieve a reasonable trade-off among these targets (Deng et al., 2025). Studies suggest that combining prediction with control optimization helps establish a closed-loop "prediction-optimization-control" framework, improving both intelligence level and operational efficiency of wastewater treatment systems (Colwell and Abolghasemi, 2024).

Although artificial intelligence has advanced rapidly in the field of wastewater treatment, existing review articles often concentrate on isolated tasks such as prediction and control, lacking a systematic synthesis and methodological summary of an integrated, closed-loop "prediction-optimization-control" framework. Meanwhile, physics-informed, data-driven modeling for engineering-ready deployment remains at an exploratory stage, and a clear technology taxonomy and applicability boundaries have yet to be established. To address these gaps, this review carries out the following work:

(1) By analyzing a large body of literature, we summarize typical water-quality prediction methods and their application scenarios, aiming to provide researchers with references for selecting appropriate methods according to their specific research needs.
(2) This paper reviews and organizes physics-informed water-quality prediction modeling approaches, covering physics-informed neural networks (PINNs), neural ordinary differential equations (neural ODEs), and physics-guided hybrid modeling, and discusses the applicable scenarios for each category of methods.
(3) This paper comprehensively surveys automation workflows in the wastewater treatment domain, with a focus on key issues for system deployment during operation, offering practitioners and researchers a closed-loop perspective and systematically elucidating the synergistic linkage mechanism of "prediction-optimization-control".

2. Methodology

This review conducted a structured literature review of English journal articles (2015–2025) retrieved from the Web of Science Core Collection. Search queries were built around "wastewater treatment plant/WWTP" and the themes of "prediction", "control", and "optimization", and refined with AI-related terms (e.g., "artificial intelligence", "machine learning", and "deep learning"); relevant studies were then screened and synthesized. Publication trends are shown in Fig. 3, and the keyword co-occurrence network is presented in Fig. 4. Based on the literature over the past decade, this review focuses on: (1) recent advances in coupling mechanistic understanding with data-driven models for wastewater quality prediction; and (2) advanced dynamic control systems and adaptive multi-objective optimization strategies for wastewater treatment processes. We also highlight a key gap in the field:

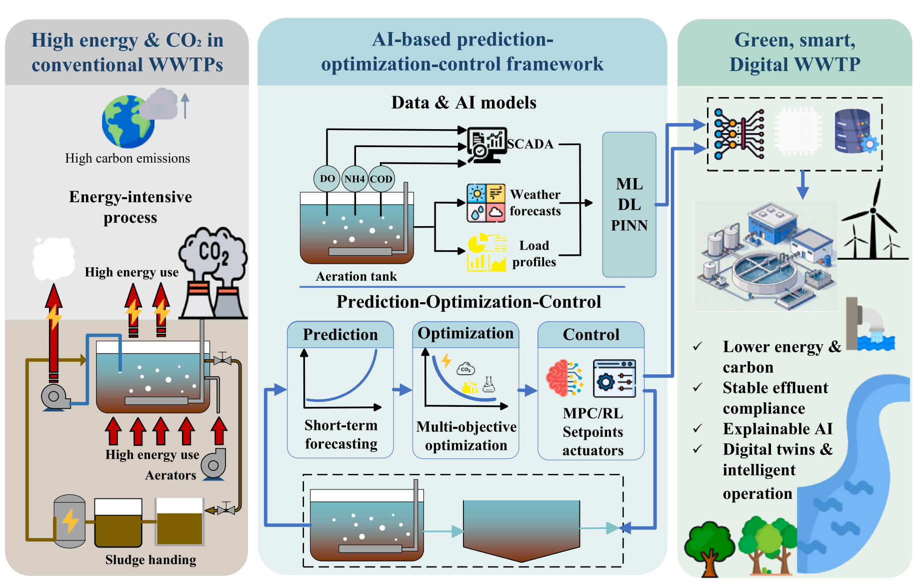

Fig. 2. AI-enabled wastewater treatment plants.

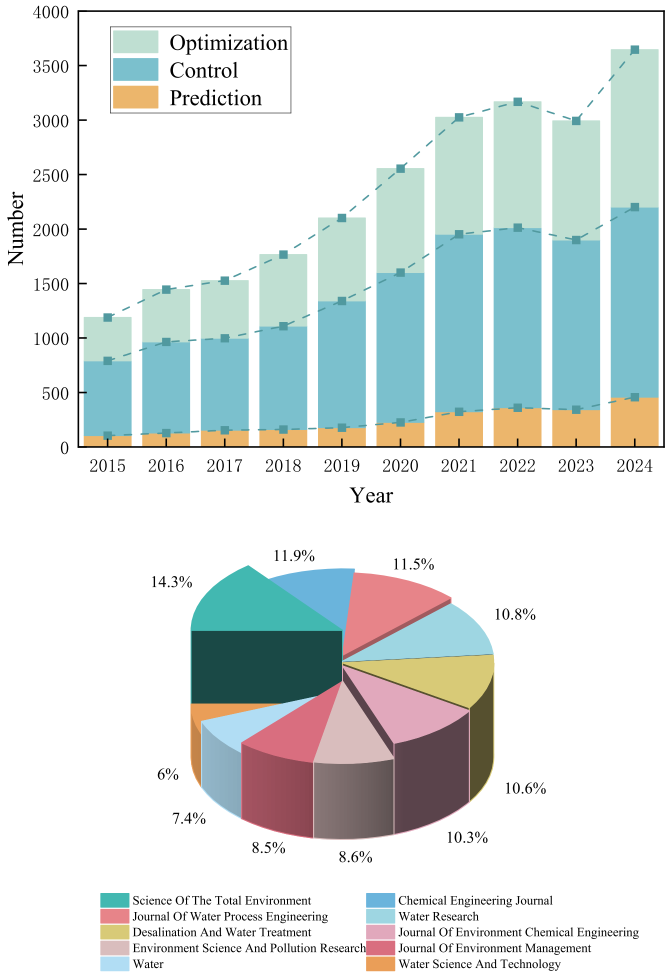

Fig. 3. (a) Literature survey over the past decade on wastewater treatment, prediction, control, and optimization. (b) Top 10 journals by number of publications in the surveyed literature.

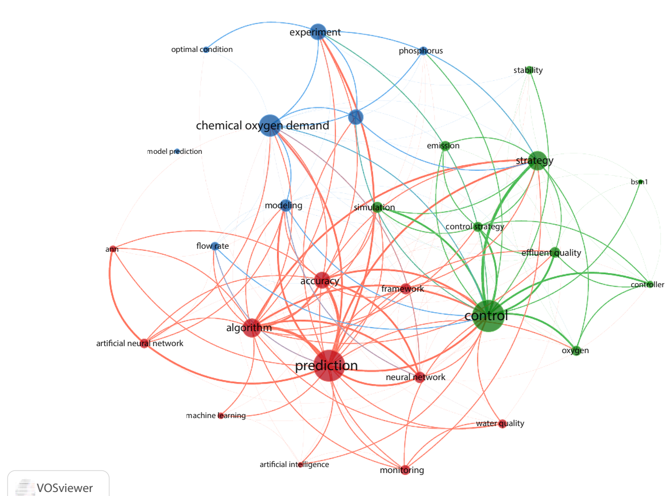

Fig. 4. Keyword co-occurrence network of the reviewed literature.

integrating activated sludge models (ASMs) with data-driven models to improve prediction accuracy and interpretability.

This review systematically examines the evolution of water quality prediction from mechanistic to physics-informed models, followed by an analysis of advanced control and multi-objective optimization strategies for enabling smart WWTPs. Fig. 5 presents the overall framework of this paper.

3. Water quality prediction models

3.1. Mechanistic water quality prediction models

Although wastewater treatment technologies have advanced, most municipal WWTPs still rely on the activated sludge process to treat domestic wastewater. This process mainly uses microbial metabolism to remove pollutants such as nitrogen and phosphorus. Activated sludge models (ASMs) are kinetic models developed based on activated sludge mechanisms, and they can quantitatively describe the rates and mass relationships of reactants and products over time (Gernaey et al., 2004). A detailed history of ASM development is summarized in Supplementary Table S1, and flowcharts of each ASM are provided in Supplementary Fig. S1.

ASMs remain widely used, and many studies apply them for process simulation and control-oriented prediction. Recent innovations in mechanistic ASM modeling mainly follow two directions. First, ASMs are simplified using stochastic differential equations or model reduction, retaining only the most influential state variables and substantially reducing computational cost (Stentoft et al., 2019). Second, the original ASM structure is extended by adding extra components and processes, such as soluble microbial products (SMP) and extracellular polymeric substances (EPS), biological phosphorus removal, and denitrifying phosphorus removal, to provide deeper mechanistic insight (Gao et al., 2017).

ASMs remain a key tool in the wastewater sector and can guide process design and operation. However, their complex reaction networks, large number of parameters, and difficulty in online calibration lead to high computational demand, which limits their use for real-time control and optimization. Moreover, many advances are validated only in short-term, single-plant studies or calibrated for individual reactors, and evidence for complex plant-sewer network coupling is still limited. Due to these constraints, research attention has increasingly shifted toward data-driven and hybrid modeling approaches.

3.2. Data-driven water quality prediction models

Compared with mechanistic models, data-driven models can automatically learn the mapping between influent conditions and effluent quality from large historical datasets, and can provide accurate predictions with limited computation. Currently, data-driven water quality prediction widely applies machine learning (ML) and deep learning (DL) to efficiently extract features from data and deliver accurate short-term forecasts, making these methods well-suited for integration into emerging smart water platforms for real-time optimization and control. In addition, ML models describe the actual behavior of reactions and processes rather than mechanisms predefined by first principles (Rios Fuck et al., 2024). Therefore, data-driven approaches can learn from experience, handle complex nonlinear patterns, and achieve more accurate prediction and inference in new scenarios, helping ensure effluent compliance while reducing energy use, operating cost, and overall resource demand (Liu et al., 2025).

3.2.1. Machine learning models for water quality prediction

ML is mainly used in wastewater treatment for short-term prediction of water quality and energy use, providing data support for operational adjustment and low-carbon operation. For water quality prediction, conventional ML is essentially supervised regression, including random forest (RF), support vector machines/regression (SVM/SVR), extreme gradient boosting (XGBoost), and artificial neural networks (ANNs).

In WWTP applications, SVM/SVR and RF are suitable for plants with small datasets, and can often achieve R^2 values above 0.90 for common indicators (Hamada et al., 2024). SVM/SVR performs well for

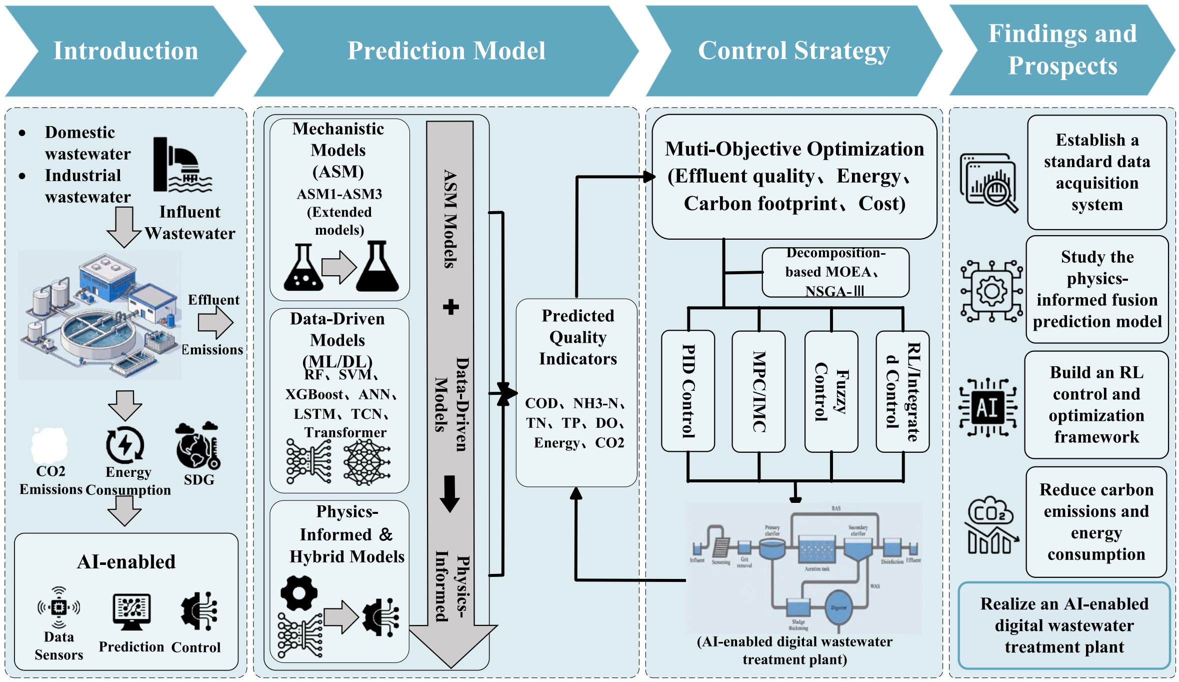

Fig. 5. Overall framework of this review.

high-dimensional nonlinear problems (i.e., many input features) with relatively low error (Li et al., 2022; Singh et al., 2022), but it is usually applied as a static regression model. RF uses a bagging structure and is robust to noise and outliers, which is useful under sensor drift and frequent data anomalies. However, RF may perform poorly when the data show strong seasonality and long-term trends (Ly et al., 2022). For medium-sized datasets where high accuracy is required, XGBoost can capture deeper nonlinear structures and improve overall performance (Xiong et al., 2025), with stable R^2 values of 0.80-0.90 for COD and total energy-related indicators. XGBoost is also strong in error control, generalization, and convergence stability due to its gradient boosting framework (Bo-Qi et al., 2025; Chen et al., 2024a). However, its higher model complexity can increase overfitting risk, which is often mitigated using optimization-based tuning strategies (Ehteram and Soltani-Gerdefaramarzi, 2025; Malik et al., 2025; Sun et al., 2025). Under large datasets and relatively stable operating conditions, ANNs can achieve $R^2 > 0.90$ for common indicators (Wang et al., 2022). For problems with strong multivariate correlations, ANNs often outperform linear and polynomial regression (Jadhav et al., 2023). They can therefore serve as effective models for soft sensing and multivariate prediction (Wang et al., 2022), and can be extended to jointly predict trace heavy metal pollutants (Matheri et al., 2021). When using real online monitoring data as inputs, SVM-based soft sensing can also show strong performance (Cechinel et al., 2024). From a closed-loop

Table 1
Comparison of practical applications of conventional machine learning for water quality prediction.

Model	Data source	Target variable	Data scale	Performance indicators	Limitations	References
RF、XGBoost、GPR、LightBoost	Gaza WWTP	TSS、BOD、COD	1 year	● R^2(TSS) = 0.932-0.979 ● R^2(COD) = 0.910-0.956 ● R^2(BOD) = 0.943-0.980	Efficiency and nonlinear fitting ability are not as good as GPR	Hamada et al. (2024)
SARIMAX、RF、SVM、GTB、ANFIS、LSTM	WWTP in Guangdong	TP	1 year	● RF(R^2) = 0.6908-0.92678	Insufficient ability to handle nonlinearity; Inability to address outliers	Ly et al. (2022)
XGBoost、RF、LightBoost	WWTP in Guangdong	COD、TEC	3 months	● R^2(COD) = 0.873 ● R^2(TEC) = 0.876	-	Chen et al. (2024a)
ANN、MLR	Ekbatan WWTP	BOD、COD	7 years	● RMSE(BOD) = 25.1 mg/L ● R^2(BOD) = 0.83 ● RMSE = 49.4 mg/L ● R^2(COD) = 0.81	More sensitive to the pH parameter	Zare Abyaneh (2014)
ANN、MLP	WWTP in Nigeria	COD	5 years	● R^2(COD) = 0.964	No simulation under actual operating conditions; Insufficient robustness	Balogun and Ogwueleka (2023)
ANN、PCA	Industrial WWTP in Aksaray	COD	9 months	● R^2(COD) = 0.9997	-	Çimen Mesutoğlu and Gök (2024)
IFFNN-LSSVM	WWTP in Beijing	BOD、NH_3-N	-	● R^2(BOD) = 0.9076 ● R^2(NH_3-N) = 0.9037	It is difficult to identify the key influencing factors	Zhu et al. (2022a)

deployment perspective, XGBoost can provide fast objective function evaluation (Al-Jamimi et al., 2024; Chen et al., 2024a), while ANNs are often embedded in multi-objective optimization as surrogate models (Mihály et al., 2022). Both are therefore easier to integrate into a "prediction-optimization-control" framework.

Overall, conventional machine learning has substantially outperformed early mechanistic models for water quality prediction, and typical applications and performance comparisons are summarized in Table 1. However, several shared limitations remain. First, these methods often rely heavily on manual feature engineering and cannot automatically extract complex spatiotemporal patterns from high-dimensional, multivariate, and coupled time-series processes in wastewater treatment (Zhu et al., 2022b).However, in the field of time prediction, some studies formulate feature selection as a binary optimization problem and propose binary meta-heuristic algorithms, such as bWWPA(Alhussan et al., 2023), AD-PRS-Guided WOA (El-kenawy et al., 2022), DTO (Takieldeen et al., 2022), these general research methods can be transferred to the field of wastewater quality prediction to mitigate the feature redundancy caused by high-dimensional inputs. Second, most studies are based on single-plant and single-season datasets; cross-season and cross-plant generalization has not been systematically validated, and performance under changing conditions and long-horizon rolling forecasts is often weak, leading to frequent retraining (Haimi et al., 2013). Third, these models generally lack interpretability and provide limited insight into underlying biochemical mechanisms. These issues become more evident under long-term drift, longer prediction horizons, and multi-objective optimization settings. With the growth of online monitoring data and computing power, deep learning methods that can learn features end-to-end and capture long-range temporal dependencies are increasingly becoming a main direction for water quality prediction.

3.2.2. Deep learning models for water quality prediction

For wastewater treatment monitoring data, which are typically high-dimensional, strongly nonlinear, and noisy, conventional machine learning relies on manual feature engineering and has limited scalability (Chen et al., 2024b). With the growth of online monitoring data and improvements in computing power, deep learning has developed rapidly. As a neural network architecture with multi-layer nonlinear mappings, deep learning can automatically extract features end-to-end, largely reducing the workload of manual feature engineering and making it suitable for complex high-dimensional nonlinear processes (Janiesch et al., 2021; Xu and Yang, 2025). Main model families include RNN/LSTM/GRU, CNN/TCN, and attention-based models such as Transformers.

In practical WWTP applications, RNN and GRU are often used in small-sample settings with hourly sampling. Because RNNs suffer from vanishing gradients, they are generally limited to short sequences for short-term prediction and are rarely used alone (Geng et al., 2022). For small datasets, GRU can train faster and sometimes perform better than LSTM, but for large datasets, it may fail to capture long-term dependencies and can still suffer from gradient vanishing (Song et al., 2020). However, a bidirectional GRU (Bi-GRU) has been reported to achieve performance comparable to, or even better than, Transformers for some indicators, suggesting that GRU-based models remain competitive in multi-task and multivariate settings (Saleh et al., 2025). For larger datasets with minute-level sampling, LSTM and TCN are more suitable. LSTM can capture longer temporal dependencies and uses gating mechanisms to alleviate the vanishing gradient issue in RNNs (Liu et al., 2023a). In contrast to the sequential computation of LSTM, TCN can enable parallel computation, improving efficiency (Bai et al., 2018; Geng et al., 2024). Moreover, TCN can be advantageous for multi-scale dynamics and multi-variable, multi-step forecasting (Li et al., 2025c). Convolutional neural networks are also used for water quality prediction, but mainly as a feature extraction front-end combined with LSTM/GRU (Shaban et al., 2024; Wan et al., 2022), or to extract features from wastewater surface images for soft sensing and multimodal sensing systems (Li et al., 2023a). For even larger datasets (e.g., sliding windows on the order of 10^5), Transformers can offer further advantages. Transformers can capture short-term shocks and long-term trends within the same layer (Ma et al., 2025a). They are well-suited for high-dimensional, long-sequence, non-stationary scenarios with clear daily cycles and seasonality and longer prediction horizons. Compared with standalone LSTM or TCN, Transformer architectures can also better model spatiotemporal dependencies in large WWTPs and multi-sensor systems (Hu et al., 2025). Attention mechanisms can focus limited computational resources on the most relevant features and time steps, helping reduce information overload under constrained computing conditions (Niu et al., 2021). For a detailed comparison of various attention mechanisms, please refer to Supplementary Material Table S2.

Deep learning models often achieve strong prediction performance and are suitable as surrogate models embedded in optimization and control frameworks (Pisa et al., 2019). However, several common limitations remain. First, deep learning is highly dependent on data quality and is sensitive to operating condition drift. Most studies still rely on single-plant, single-season datasets, with limited cross-plant validation. Second, many studies use extensive trial-and-error hyperparameter tuning without incorporating process knowledge such as hydraulic retention time or diurnal loading cycles, and instead rely mainly on grid search (Yuan and Li, 2025). However, in the time-series forecasting literature, some studies have employed meta-heuristic optimization to jointly tune model hyperparameters and weighting coefficients, providing a general strategy to reduce reliance on manual trial-and-error (El-kenawy et al., 2023). Third, a considerable number of studies lack systematic ablation experiments, and model complexity is sometimes not well matched to the available data size, requiring intelligent optimization and distributed computing to meet efficiency needs (Bi et al., 2025). Fourth, closed-loop integration with process control and optimization is still limited. Few studies have embedded Transformer-based predictors into model predictive control or reinforcement learning frameworks, and even when integration is reported, full-scale validation for practical deployment is often missing. Finally, model interpretability remains insufficient, and the plausibility of predictions is not always well supported.

3.2.3. Hybrid models

Ensemble approaches can combine the strong feature extraction of ML/DL models with enhanced sequence modeling, and thus build hybrid models that outperform any single model, improving the efficiency of water quality prediction. Hybrid models include ensembles of DL and ML, ensembles of multiple DL models, and integrations of mechanistic and data-driven models. These hybrid models can capture different aspects of the data, tracking temporal changes while identifying key drivers that affect treatment performance (Zamfir et al., 2025). Compared with single-model approaches, this dual capability helps address the noisy and complex nature of water quality data (Anandan and Sundaram, 2024).

Within a purely data-driven framework, a single recurrent network or a single tree-based model often cannot simultaneously capture temporal dependence and nonlinear regression. This has motivated hybrid schemes that connect DL and ML models in series or in parallel. In these schemes, DL is used to capture temporal dependencies, while ML (e.g., tree-based or kernel methods) improves nonlinear fitting and is often used for residual correction (Xiong et al., 2025). Another type of hybrid model couples components within a DL architecture, such as combining CNN/TCN with LSTM and attention modules. Here, convolutional blocks extract local and multi-scale features, and recurrent networks capture long-term dependencies (Xie et al., 2024). A third type integrates mechanistic and data-driven models, where the mechanistic model enforces mass conservation and process dynamics, while the data-driven part fits residuals to improve accuracy. Compared with black-box models, these hybrids offer better interpretability (Xu et al., 2024). In practice, hybrid frameworks are particularly useful in high-complexity scenarios such as lake watershed systems(Guo et al., 2024), multimodal soft sensing (Mai et al., 2023), and multi-indicator joint prediction (Li et al., 2025c). For small-data scenarios such as single-indicator short-term forecasting, a single model is often more appropriate and cost-effective.

Overall, although hybrid frameworks can improve prediction accuracy, they also have notable limitations. A common issue is a mismatch between model complexity and data size. For example, complex architectures such as CNN-LSTM-attention may be trained on only hundreds to thousands of samples, which increases the risk of overfitting (Liu et al., 2024a). In addition, many studies do not provide ablation experiments for rigorous comparison.

3.2.4. Limitations and future directions

This section reviews data-driven approaches for wastewater quality prediction, with an overview of commonly used ML and DL models in this field (Table 2). Compared with traditional mechanistic models, data-driven models show clear advantages. However, several limitations remain. First, water quality data are often of insufficient quality and limited availability. They can be affected by seasonal variation, sensor drift, and irregular sampling intervals, which introduce additional noise and uncertainty into modeling and prediction (Lokman et al., 2025). Second, most current data-driven models focus on short-term forecasting, while their accuracy for long-horizon prediction is still limited. This is partly because the influence of historical data on far-future states weakens, and because multiple uncertain factors (e.g., extreme weather and seasonality) affect wastewater systems. Finally, interpretability remains a major challenge. Although ML and DL can provide accurate predictions, their internal logic is often not transparent. This “black-box” nature reduces acceptance among researchers and practitioners who require mechanistic plausibility (Lokman et al., 2025). Therefore, coupling ML/DL with physics-informed networks and embedding physical mechanisms as model constraints to ensure physically consistent predictions is an important direction for future development.

3.3. Physics-informed water quality prediction models

3.3.1. Physics-informed neural networks (PINNs)

Physics-informed neural networks are interpretable models that combine physical information (e.g., physical principles or governing equations) with deep learning. The most common PINN approach is to embed physical laws (such as partial differential equations) into the loss function, so the model is trained under physical constraints. This can improve generalization and interpretability, and is especially useful when data are scarce or noisy. The soft-constraint structure of a PINN is illustrated in Fig. 6. Such physics-data integration can not only improve prediction performance but also help mitigate the “black-box” nature of neural network training (Koksal and Aydin, 2025).

The most common approach is a soft-constraint scheme based on a customized dual-objective function, and the detailed derivation is provided in Supplementary Material S3. Other approaches include hard-constraint integration by modifying the network architecture or output layer to embed physical information.

First, hard-constraint PINNs differ from soft-constraint PINNs in that boundary conditions are not penalized in the loss. Instead, physical constraints are enforced through the network structure or output transformation. In this way, physical laws are “written into” the model, and the network cannot violate the physical conditions regardless of the training process. Related methods can be found in (Lu et al., 2021).

Second, mechanistic-neural hybrid models combine mechanistic models with neural networks and are closer to engineering deployment in wastewater applications. For example, an ASM can be used to generate a mechanistic prediction, and a neural network is then trained to learn the residuals or correction factors between the mechanistic outputs and measured data. Alternatively, poorly modeled kinetic terms can be represented by neural networks and used as components of the governing equations, leading to joint fitting of mechanistic equations

Table 2
Studies on other data-driven models for water quality prediction.

Model	Experimental subject	Prediction Indicators	Experimental Results	References
LSTM	Nanjing AAO WWTP	DO、COD	● R^2(DO) = 0.97 ● R^2(COD) = 0.87	Ma et al. (2025b)
	Shenzhen WWTP	COD、TN、TP	● R^2(COD) = 0.914 ● R^2(TN) = 0.924 ● R^2(TP) = 0.904	Zhang et al. (2023)
	Denmark Avedøre WWTP	N_2O	● $R^2 > 0.94$ ● RMSE reduced by 64%	Hwangbo et al. (2021)
LSTM + ARIMA	WWTP	NH4+、NO3-	● $R^2 = 0.95$	Seshan et al. (2024)
TL-LSTM	Sichuan AAO WWTP	COD	● RMSE = 0.627 mg/L ● $R^2 = 0.811$	Yu et al. (2024)
Bi-LSTM + CNN	Chongqing AAO WWTP	COD、NH3-N	● Performance improved by 5%	Zeng et al. (2021)
LSTM + MLP	WWTP in Southern China	COD、NH3-N	● MSE(COD、NH3-N) = 2.129、0.706 ● MAE(COD、NH3-N) = 1.198、0.650 ● R^2(COD、NH3-N) = 0.799、0.824	Wei et al. (2023)
DNN + LSTM	ShanXi Provence WWTP	Polymer Dosage	● R^2 = 92.7%	Ma et al. (2024)
Bi-LSTM + GA + PCA	WWTP	NH4-N、PO4-P、COD	● R^2 = 0.93-0.98	Kazadi Mbamba and Batstone (2023)
TCN + Bi-LSTM + MultiHead Attention	Chemical Industry WWTP	BOD、COD、NH3-N	● $R^2 = 0.9638$ ● MSE = 0.0008	(S et al., 2025)
ANFIS + MPC	WWTP in Modena, Italy	TN	● Energy consumption reduced	Bernardelli et al. (2020)
RBNN + TDNN	WWTPs	NO^{3-}、DO	● MAPE = 1.27-2.57	Mihály et al. (2022)
AFAE + ACNN	WWTPs	TSS、COD、DO	● R^2(DO) = 0.9909 ● R^2(TSS、COD) = 0.9951、0.9894	Ba-Alawi and Kim (2025)
GCN-CNNGA	WWTP in eastern China	COD、NH3-N、TP、TN	● An improvement of more than 16.7% in key indicators such as RMSE, MAE, and R^2	Cao et al. (2024)
SVR-GA	Southern Tehran Municipal WWTP	Biogas Production	● $R^2 = 0.773$	(Farzin et al., 2024)]
TCANRF	A WWTP in the Taihu Lake Basin	COD、TN、TP	● MAPE reduced by 20%	Li et al. (2025a)
BN	Iran, Australia, Spain WWTP	TSS、TP	-	Roohi et al. (2024)

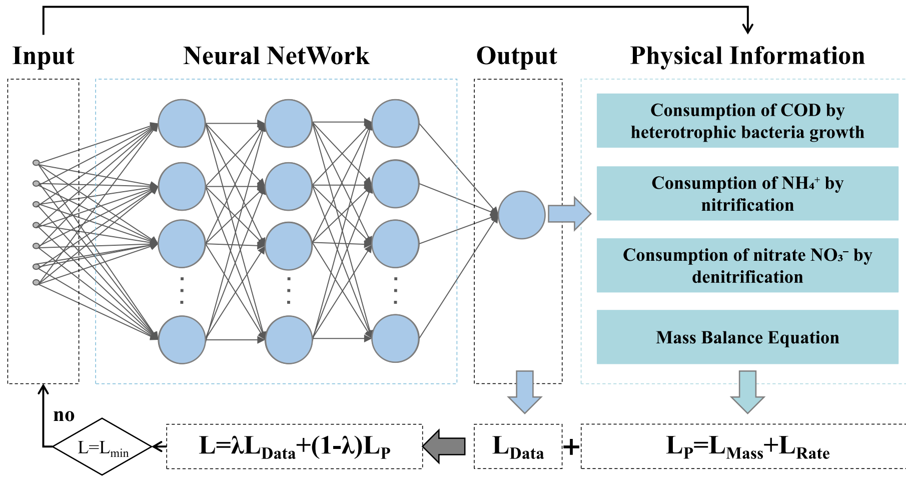

Fig. 6. Soft-constraint structure of a PINN(Garakani and Chew, 2025).

and neural differential equations.

Finally, structure or graph-constrained models align learning with industrial topology. Because WWTPs have a clear process topology, graph neural networks or modular networks can be built to represent units and recycle flows, propagate states across nodes, and incorporate physical operators such as flow conservation and concentration mixing into the message passing rules. Although this is not a standard PINN, the network structure itself reflects the process configuration. Representative methods are described in (Wu et al., 2024).

3.3.2. Physics-informed data-driven model

Wastewater treatment processes inherently follow flow and mass conservation, effluent boundary conditions, ASM kinetics, and a fixed plant topology (wastewater–sludge streams and recycles). These mechanisms can be used as physical knowledge in PINNs to prevent violations of basic process constraints, thereby improving the credibility of predictions and facilitating subsequent integration with optimization and control. By embedding physical knowledge as a regularizer, PINNs can also reduce overfitting and improve training efficiency, and have attracted broad attention in process modeling (Shaw et al., 2024). Therefore, PINNs provide a promising way to couple ASM-based mechanisms with data-driven models, although only a limited number of studies have explored this direction.

In existing work, the most common strategy is to embed ASM-based mechanisms as soft constraints in the neural network loss. Koksal et al. modeled dissolved oxygen (DO) and chemical oxygen demand (COD) in an industrial WWTP using a hybrid PINN and data-driven approach. They extracted ASM-based expressions for DO and COD, where COD concentration was approximated by the rate of readily biodegradable substrate. This expression was used as a physics-based soft constraint with a tunable weight. Using 100 days of online data for long-term validation, the physics-informed LSTM reduced the MSE for DO prediction by 20%, and the physics-informed GRU reduced the MSE for COD prediction by 11%. The model also remained functional 1.5 years after the last training, indicating that PINNs can improve both accuracy and long-term stability when an exact mechanistic model is unavailable (Koksal et al., 2024). The same group applied a PINN framework to the industrial wastewater treatment unit of the Tüpraş İzmit refinery and built a dual-objective loss guided by physical information. The PINN markedly improved prediction performance, with an 82% reduction in MSE for DO prediction (Asrav et al., 2023). During training, careful normalization of different loss terms is critical; dynamic or adaptive weighting can help tune the physics weight and avoid non-convergence.

The soft-constraint approach above is widely used because it does not require large datasets and has a clear physical meaning. Other strategies can also integrate physical information. For example, one study focused on key water treatment units (e.g., CSTRs, activated sludge reactors, and fixed-bed adsorption reactors) and constructed PINNs by embedding physical laws into the network architecture (Li et al., 2024). While hard constraints offer strong interpretability, they typically require higher-quality data and are harder to train and converge than soft constraints. Continuous-time neural frameworks based on neural ODEs or hybrid ODEs have also been explored (Song et al., 2025), such as embedding ASM stoichiometric relations into neural networks to build hybrid ordinary differential equation models (Zhao et al., 2025). These methods may be suitable for full-scale plants with high-quality data, but training can be unstable for stiff systems and sensitive to drift and noise. For whole-plant, multi-unit, multi-sensor settings, PINNs can be combined with graph neural networks (GNNs) to represent plant topology (Alevizos et al., 2025), although online coupling with control remains challenging. Other extensions include combining transfer learning with PINNs (Koksal and Aydin, 2025). Further research is still needed to develop interpretable models for wastewater treatment by deeper integration of PINNs and neural networks.

PINNs, integrated with mechanistic models and data-driven approaches, offer a research paradigm that combines theory-based mechanistic constraints with efficient computation for complex models, thereby improving robustness and generalization. A comparison of different physics-informed fusion methods is provided in Supplementary Table S3.

However, several challenges remain. First, training PINNs often requires carefully designed loss functions and a balanced weighting between physics-based terms and data-driven terms, which increases computational cost and slows convergence. Second, full ASM models are complex and high-dimensional, consisting of multiple coupled and nonlinear ordinary differential equations (ODEs) with stiff dynamics. Embedding an entire ASM family as soft constraints can make training extremely difficult and may lead to non-physical convergence. Third, most existing studies are still limited to numerical experiments or small-scale process validations. Practical deployment of PINNs in engineering settings still requires improvements in computational efficiency, real-time prediction, and cross-plant generalization. Their scalability and adaptability in full-scale operation need to be validated through more extensive empirical studies.

4. Control and optimization strategies

Water quality prediction is a prerequisite for intelligent decision-making and operational optimization in WWTPs. Only by integrating prediction results with control and optimization strategies can plant operation be effectively guided at the system level. As shown in Fig. 7,

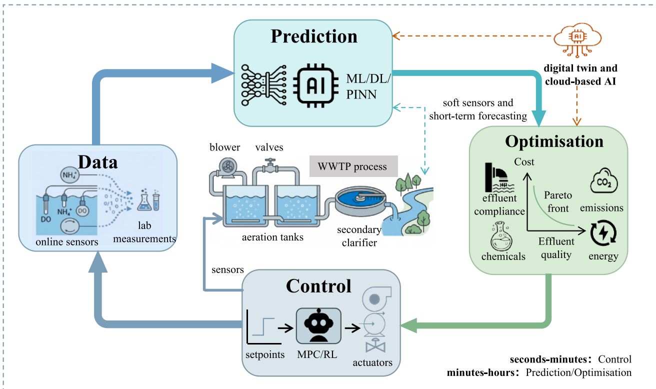

Fig. 7. Closed-loop framework of prediction–optimization–control.

the predictor uses historical data to forecast effluent quality over the next several time steps. The optimization layer then computes an optimal control policy by balancing multiple objectives (e.g., effluent quality, energy use, and cost). Finally, the control layer applies only the control actions for the current time step in a receding-horizon manner. Intelligent control can replace operators to automatically regulate overall plant operation, reducing the risk of increased energy consumption caused by inaccurate judgments and improper actions. Because aeration can account for 60% of the total plant energy consumption (He et al., 2019), most control and optimization studies focus on key variables such as dissolved oxygen and aeration intensity.

4.1. Advanced model-based intelligent control

In current aeration and DO automation systems, proportional-integral-derivative (PID) control remains the dominant solution because of its simple structure and extensive engineering experience (Man et al., 2018). Recent improvements mainly include adaptive PID tuning, PID parameter tuning using intelligent optimization algorithms, and modifications of the PID structure (Aparna and Swarnalatha, 2023; Du et al., 2018; Li et al., 2023b). However, as aeration systems become more nonlinear and time-varying, PID alone cannot meet the needs of plant-wide multivariable and multi-objective optimization. It is more suitable as a low-level actuator controller within a future "prediction-optimization-control" architecture, tracking setpoints generated by upper-layer optimization.

To overcome the limitations of PID in multivariable coupling and multi-objective trade-offs, researchers have proposed advanced control strategies based on data-driven models. These strategies extend the control objective from steady-state error in a single loop to integrated trade-offs among coupled variables, effluent quality, and cost. Representative methods include fuzzy logic control (FLC), model predictive control (MPC), and internal model control (IMC) (Gu et al., 2023). FLC can leverage expert knowledge when accurate mechanistic models are unavailable and is robust to parameter variations, making it suitable for nonlinear, time-varying, and partially modeled systems (Ribes et al., 2023). However, it is difficult for FLC to represent multiple objectives (e.g., energy use and EQI) within a single rule base. MPC is well-suited for multivariable systems with constraints, because multiple objectives such as energy, effluent quality, and cost can be formulated within one objective function (Wang et al., 2025b). MPC can also be combined with data-driven models to improve the prediction of influent disturbances and reduce online optimization burden (Protoulis et al., 2025). Importantly, an ANFIS-based MPC has been deployed in a large municipal WWTP (500,000 population equivalent), demonstrating engineering feasibility (Bernardelli et al., 2020). Nevertheless, most studies are still limited to simulation platforms, with a lack of long-term full-scale validation. In addition, carbon emissions and sludge-line energy use have not been systematically included as control objectives. IMC has a simpler structure and is often considered a compromise between MPC and PID. It can handle time delays better than empirically tuned PID (Pisa et al., 2020), and is more suitable for multivariable settings than decoupled PID, including water-sludge line coupling and coordinated aeration–recycle control (Rojas et al., 2016) However, IMC still depends on reliable models and may involve higher maintenance costs.

To balance model reliability and nonlinear representation, increasing efforts are being made to integrate neural networks with MPC, IMC, or fuzzy control, such as dynamic multi-objective immune optimization control (Li et al., 2021a)and hierarchical nonlinear adaptive control (Piotrowski et al., 2023). Compared with standalone neural network controllers, these integrated approaches introduce state and input constraints into control optimization, offering more robust compensation for strong nonlinearity and time-varying behavior in WWTPs (Ren et al., 2022; Schimperna et al., 2024). In addition, reinforcement learning (RL) can learn control policies directly through trial-and-error feedback, and has shown potential to reduce both violation risk and energy use in aeration control, outperforming conventional PID in some studies (Logan et al., 2025). However, RL training is highly sensitive to reward design, exploration strategies, and the quality of simulation environments. Performance may degrade over long training horizons, and long-term stability and safety remain major barriers for engineering deployment. Therefore, RL may be more suitable as an upper-layer setpoint recommender or policy search tool, while the lower layer still relies on verifiable constraint-based controllers.

Overall, MPC performs well for constrained multivariable optimization, IMC provides a lightweight model-based option between PID and MPC, FLC and its intelligent variants can effectively use expert knowledge when mechanistic models are unavailable, and integrated control aims to coordinate multiple objectives (e.g., water and sludge lines, energy use, and effluent quality) within a unified framework. Table 3 summarizes the control strategies and structures of different algorithms. However, several limitations remain. First, most studies are still limited to simulation or short-term single-plant applications, with little systematic evaluation of cross-season and cross-plant generalization. Model uncertainty, sensor faults, and equipment constraints are also not fully considered in terms of control performance and safety. Second, advanced control is often studied separately from process prediction and optimization, and truly integrated closed-loop frameworks linking "soft sensing-prediction-multi-objective optimization-control" are still rare. Future work should systematically compare control strategies on standardized simulation platforms and multi-source full-scale datasets, and clarify how advanced control can be embedded into closed-loop architectures with demonstrated engineering value.

4.2. Multi-objective intelligent optimization strategies

Wastewater treatment systems include both the water line and sludge line and exhibit strong nonlinearity and fluctuations, making it difficult for control systems to maintain operation at an optimal setpoint (He et al., 2025). A WWTP can be formulated as a dynamic optimization problem that seeks trade-offs among multiple objectives such as energy use, chemical and sludge disposal costs, and greenhouse gas emissions under the constraint of effluent compliance (Liu et al., 2024b). Key decision variables typically include DO setpoints in aeration tanks, internal/external recycle ratios, external carbon/chemical dosing rates, and sludge age. Objective functions are often defined using different combinations of effluent quality indices (EQI) or risk-based indicators, energy use, chemical and sludge disposal costs, and greenhouse gas emissions.

Multi-objective optimization studies can be broadly grouped into three categories. The first focuses on the trade-off between effluent quality and energy use. These studies often use DO setpoints and recycle ratios as decision variables, and take effluent quality and energy as objectives on BSM1/2 simulation platforms. Multi-objective algorithms are used to construct Pareto fronts and identify a set of optimal operating

Table 3
Studies on control algorithms for wastewater treatment processes.

Method	Purpose	Control Strategy	Result	References
GA-PID	Biomass–influent ratio control	● GA; ● Small Gain Theorem; ● H∞ Criterion	Robust PID tuning under uncertainty	(Xing-hong et al., 2015)
PPID	DO concentration & nitrate level control	● A network-based predictive PID controller; ● The Truetime network simulator is used to test	Superior DO control (vs. PI, BSM1); tolerance to data loss	Vásquez (2022)
BP-PID Fuzzy-PID	DO concentration control	● BP neural networks ● Fuzzy logic rules	Improved energy-consumption control; strong disturbance rejection	(Shen et al., 2016)
MARS-MPC	Carbon source dosing optimization	● Integrate with the Soft Sensor Based on Multivariate Adaptive Regression Splines	34.3% reduction in carbon consumption; stable nitrogen removal	Yang et al. (2025)
NNMPC	DO concentration control	● Dynamic Setpoint Adjustment ● Constrained Neural Network Model Predictive Control	Improved effluent quality; reduced overall energy consumption; robust to weather variability	Sadeghassadi et al. (2018)
ACWA	DO concentration control	● Introduce a weighted action-value function	Better performance than conventional PID and basic ADP	Wang et al. (2024)
GD-MOEA/D	Energy use & effluent quality	● Based on the Grid Dynamic Multi-Objective Evolutionary Decomposition Algorithm	Reduced energy consumption; improved effluent quality	Xie et al. (2022)
BO-RL	Operating cost & energy consumption	● Combine Reinforcement Learning with Bayesian Optimization	46% reduction in operating costs; 12% reduction in energy consumption	Zhu et al. (2025)
EAGSSA-RESN	BOD_5、COD	● Elite Chaotic Opposition-Based Learning ● Nonlinear Adaptive Weight Factors	Lower RMSE than SSA and RESN; RMSE reductions of 25.41%, 7.45%, 6.34%, and 24.97%	Liu et al. (2024c)

points under simulation conditions (Aparna and Swarnalatha, 2023; Liu and Jiang, 2024; Xie et al., 2022). However, many results remain offline and limited to single operating scenarios, with insufficient robustness analysis under extreme rainfall or abrupt influent changes. A practical direction is to convert Pareto solutions into an "operating point library" and enable online selection and updating through condition recognition or rolling optimization. The second category adds carbon emissions as an explicit objective. Many studies use NSGA-II and incorporate greenhouse gas emissions and/or life cycle assessment (LCA) into the objective function to obtain three-objective Pareto solution sets (Huixin et al., 2023; Liao et al., 2024). However, carbon accounting boundaries vary across studies, including indirect emissions from electricity use only,

Table 4
Other studies on multi-objective optimization algorithms.

Algorithm	Optimization objectives	Control Strategy	Performance Metrics	Advantage	Limitation	References
MSAMODE	OCI + EQI	LADRC + ESO	-	Enhance the convergence performance of Pareto solutions	● Insufficient precision ● No practical verification	Liu et al. (2023b)
NSMOCS + MOGA	OCI + EQI + NH_4-N	PI	The EQI decreased by 0.77% to 1.3%; The OCI decreased by 0.63% to 0.9%.	Enhance optimization efficiency and solution quality	● Complex calculation ● Single control variable	Aparna and Swarnalatha (2023)
NSGA-II + TOPSIS	EQ + EC	-	The EQ optimization rate is 19.377%, and the EC optimization rate is 1.552%.	Enhance model interpretability using the SHAP method	● The impact of hyperparameters is not considered ● The scope of feature selection is limited	Liu et al. (2024b)
ICEEMDAN + IDBO + MOALO + IISRBS	EC + EQ	LADRC	-	Enhancement of the algorithm's convergence speed	● Insufficient performance ● No verification under multiple operating conditions	Liu and Jiang (2024)
DMOPSO-CD	EC + EQ	The underlying controller based on SOFNN	EC decreased by 6.28% to 7.27%.	Avoid premature convergence of the algorithm	● The calculation is complex ● No undergone practical verification	Dai et al. (2023b)
ANSGA-III	AEQ + OCI + TV	-	AEQ decreased by 13%, while OCI dropped from 21,023 kWh d^{-1} to 20,226 kWh d^{-1}. TV decreased from 17,065 m^3 to 16,530 m^3.	Reliability has been improved	● Complex structure ● Has limitations in addressing complex problems	Dai et al. (2023a)
RL:TD3	DO + IMLR + RAS + WAS Environment and Cost	RL Agent	The performance indicator has increased by 8.3%.	Multi-action Control and Unified Control Strategy	● The results exhibit variability ● Insufficient accuracy of the reward function	Croll et al. (2024)
Goal Programming Approach	Environment and Cost	-	The cost target value is $58,566.53, the environmental target value is 464,910 kg CO_2-eq, and the efficiency score is 75.4%.	Synergistic Optimization of Sewage Treatment and Sludge Treatment	● Limitations in assumptions and data ● No practical verification	Caligan et al. (2022)
RL + BO	COD + Cost + Energy Consumption	Control Based on Traditional Rules and RL	The operating cost has decreased by 46%, the energy consumption has dropped by 12%, and the effluent quality meets the discharge standards.	RL and BO are integrated	● Cannot capture the complexity of real-world scenarios ● The results exhibit variability	Zhu et al. (2025)

direct process emissions, or plant-wide LCA. In addition, sludge-line energy use and indirect emissions are often omitted. As a result, results are not directly comparable across studies. Future work should clearly report system boundaries, emission factor sources, and whether the sludge line is included. The third category aims to couple multi-objective optimization with control. One approach combines upper-layer optimization with lower-layer control: an upper-level multi-objective optimizer uses short-term predictions to compute optimal setpoints, and a lower-level PID or MPC controller tracks these setpoints to achieve trade-offs among objectives (Cai et al., 2025). Another approach embeds a multi-objective cost function directly into the MPC design to support multi-objective decision-making under constraints (Han et al., 2022). However, systematic reporting of real-time performance and closed-loop stability is still limited, and prediction uncertainty should be included in closed-loop evaluation.

Overall, multi-objective optimization in WWTPs has demonstrated the feasibility of identifying optimal setpoints on benchmark models (BSM1/2) and in some full-scale cases (Table 4). Yet several gaps remain: (1) most studies rely on single-plant, short-term, or idealized mechanistic settings, with limited robustness assessment under extreme events and long-term operation; (2) carbon optimization often uses simplified linear emission factors, with few studies performing plant-wide LCA covering both water and sludge lines; and (3) optimization and control are often coupled offline, and standardized closed-loop solutions that can be transferred to full-scale practice are still lacking. Future work should move toward engineering deployment by adapting multi-objective weights to operating conditions, including sludge-line optimization, and enabling low-carbon optimization under a full plant boundary.

4.3. Building a closed-loop framework

We frame smart WWTP operation as a deployable closed-loop "monitor-predict-optimize–control" architecture. Online sensor streams and laboratory measurements are assimilated to update plant states, which then drive data-driven and mechanistic–data hybrid (or physics-informed) models to generate short-horizon forecasts of key variables and operational risks. These forecasts are passed to a multi-objective optimizer to compute feasible operating setpoints under hard process constraints and a consistent plant-wide system boundary (e.g., effluent compliance, energy use, and carbon impacts, including the sludge line/LCA boundary when relevant). The resulting setpoints are implemented by validated lower-level controllers (e.g., PID or MPC) to ensure stability, safety, and real-time execution. Measured performance is finally fed back to recalibrate models and revise decisions on a rolling (receding-horizon) basis, improving robustness to disturbances and drift while remaining practical for full-scale deployment.

In addition, the practical deployment of a closed-loop framework is constrained not only by engineering conditions but also by ethical and governance considerations. Missing data, drift, and transmission anomalies in online streams, as well as systematic biases introduced by differences in operating regimes and labeling practices across plants, may undermine reliability under cross-plant transfer and extreme conditions and thereby introduce decision-making risks. Since closed-loop control directly affects energy consumption, effluent quality, and operational safety, the model's scope of applicability, accountability, and audit mechanisms should be clearly defined, and human-in-the-loop supervision and fail-safe fallback strategies should be adopted. Meanwhile, computational scalability and real-time requirements should be considered; integration pathways with existing control systems should be clarified; essential latency and resource-usage metrics should be reported; and traceable monitoring and update strategies should be established to support trustworthy and reproducible engineering applications.

5. Challenges, opportunities, and future perspectives

Overall, the literature indicates that AI is reshaping the operational paradigm of WWTPs. However, there is still a clear gap between current research and deployable, trustworthy smart WWTPs, mainly in four aspects: data foundations, model trustworthiness, and closed-loop control.

(1) Insufficient data quality and standardization remain major challenges, and many existing studies rely on small-scale, single-source datasets. Wastewater treatment plant data are typically high-frequency, long-term, and multi-source/heterogeneous. Online sensors are susceptible to environmental disturbances, sensor accuracy limitations, and data transmission failures, which can lead to missing records. In addition, operating conditions and influent water-quality characteristics differ across plants, and data standards are often inconsistent, limiting cross-plant generalization. Furthermore, the widespread use of small and single-source datasets reduces model robustness under unseen conditions and weakens overall generalizability.

(2) High overfitting risk and insufficient methodological transparency. Many existing studies rely on a single train–test split and randomly partition time-series data or use datasets covering only a short time span, which can cause information leakage and thus increase the risk of overfitting. In addition, some studies lack sensitivity analyses and ablation experiments, making it difficult to quantify the actual contribution of each module to performance improvements and reducing methodological transparency and reproducibility.

(3) It is difficult to fully embed mechanistic models and balance physical weights. Incorporating the entire ASM as a soft constraint will complicate the model loss function, making training difficult to converge and even leading to non-physical convergence. Moreover, fine parameter tuning is required for the weights of physical and data terms, resulting in high computational costs. In addition, few studies have integrated physics-informed hybrid models into the optimization and control framework, leading to insufficient deployability.

(4) Limited real-time performance and incomplete control–optimization frameworks. Conventional or simple controllers cannot respond effectively to rapid water quality fluctuations, while advanced intelligent controllers may require heavy computation that limits real-time industrial deployment. Moreover, many studies do not integrate plant-wide boundaries—such as life-cycle carbon footprint modeling and the sludge line—into optimization and control, which reduces practical relevance and deployment potential.

Intelligent transformation is an important long-term trend for wastewater treatment, but it requires addressing these challenges (Fig. 8). Future work may focus on four directions:

(1) Establish cross-plant data and benchmarking systems. Advanced sensing and data augmentation, together with edge–cloud collaboration, can improve data quantity, quality, timeliness, and cross-plant usability, supporting robust and accurate modeling (Shirkoohi et al., 2022). Clarify data standardization principles—such as variable definitions, units, sampling frequency, quality control, and missing-data handling—build publicly available and reusable benchmark datasets as well as high-quality labeled datasets, and establish unified evaluation and cross-plant validation protocols, thereby improving model generality and transferability. In future studies, it is recommended to use large-scale, multi-site datasets whenever possible to enhance model robustness and generalization under unseen operating conditions.

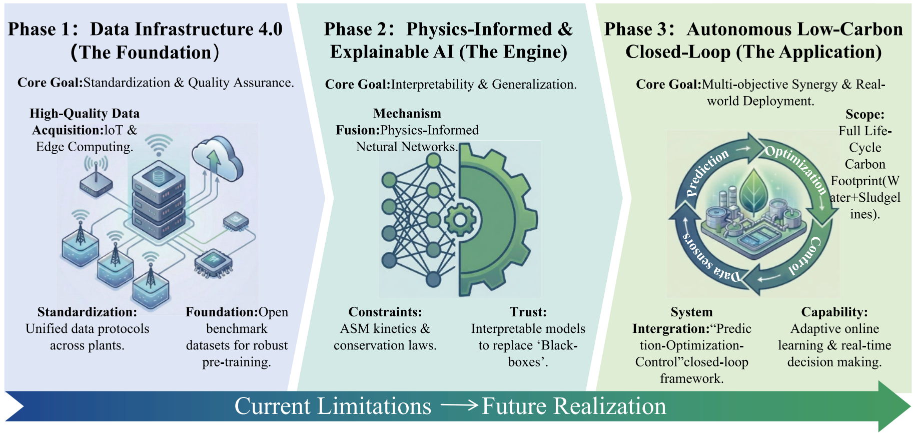

Fig. 8. Future roadmap for AI in wastewater treatment.

(2) Adopt more appropriate validation strategies to reduce overfitting risk and improve methodological transparency. Future studies should combine k-fold cross-validation/repeated experiments/nested validation with time-series blocked splits, and conduct evaluations across seasons and operating conditions as well as under abnormal scenarios such as sensor missingness, drift, and extreme influent events, to enhance robustness and external validity; meanwhile, variability ranges and confidence intervals based on multiple random seeds should be reported. In addition, sensitivity analyses and ablation studies are recommended to quantify the contributions of key hyperparameters, physics-loss weights, and critical modules, thereby strengthening methodological transparency and reproducibility.
(3) Develop physics-informed and interpretable AI models. During the training of physics-informed hybrid models, normalization should be performed on different loss terms, and adaptive weighting or staged training strategies should be adopted to mitigate the challenge of weight adjustment. In addition, future research needs to integrate the topological relationships of WWTPs into PINNs. Meanwhile, it is essential to implement the practical engineering deployment of physics-informed hybrid models, and conduct robust evaluations against indicators such as cross-seasonal variations, cross-plant transferability, extreme influent events, and sensor anomalies.
(4) Enable a closed-loop "prediction-optimization-control" framework with stronger adaptability. Prediction, optimization, and control should be studied as an integrated chain to build a unified closed-loop framework. Plant-wide boundaries, including life-cycle carbon footprint models and the sludge line, should be incorporated, followed by full-scale validation to accelerate engineering deployment. Adaptive optimization and control models with continuous monitoring should be developed to detect operating changes, update strategies automatically, correct model drift and performance degradation, and improve performance through online learning under real operating conditions (Chen et al., 2025).

In addition, emerging technologies such as the Internet of Things (IoT) and RL serve as important driving forces for the intelligent development of wastewater treatment, enabling efficient real-time monitoring and data collection. Therefore, the future intelligent transformation of WWTPs is expected to advance by virtue of in-depth integration with AI technology. However, the success of this transformation hinges on the parallel advancement of data infrastructure construction, the development of trustworthy models, and full-scale engineering validation.

6. Conclusion

This review systematically summarizes research over the past decade on AI applications in WWTPs, with a focus on achieving low-carbon operation through an integrated "prediction-optimization-control" approach. Different from previous reviews, we highlight the need to embed physical knowledge into AI models and to build closed-loop frameworks that are both interpretable and generalizable. The main findings indicate that physics-informed neural networks, hybrid modeling, and multi-objective reinforcement learning have strong potential to support sustainable WWTP operation.

To enable trustworthy AI-enabled WWTPs, future research should prioritize the following directions:

(1) establish high-quality cross-plant and cross-climate data infrastructure and open benchmark datasets to evaluate the generalization and robustness of AI models under real operating conditions;
(2) advance physics-informed and interpretable AI by combining ASMs, PINNs, and uncertainty quantification to improve confidence in engineering decision-making;
(3) develop dynamic, life-cycle carbon footprint optimization models that include the sludge line based on water quality forecasts, and couple multi-objective optimization with control strategies to form a closed-loop framework.

Addressing these challenges will facilitate deeper integration of intelligent wastewater treatment with Industry 4.0 technologies and accelerate the transition toward smarter and greener WWTPs.

CRediT authorship contribution statement

Li Yiting: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Zhang Hongpeng:** Writing – review & editing, Supervision. **Zhang Peng:** Writing – review & editing, Supervision, Formal analysis. **Wang Yin:** Writing – review & editing, Resources, Project administration. **Hao Yiming:** Writing – review & editing, Visualization. **Wang Xinran:** Writing – review & editing, Visualization. **Guo Wenxuan:** Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported by the National Key Research and Development Program of China (2024YFC3810201).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvman.2026.128949.

Data availability

No data was used for the research described in the article.

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